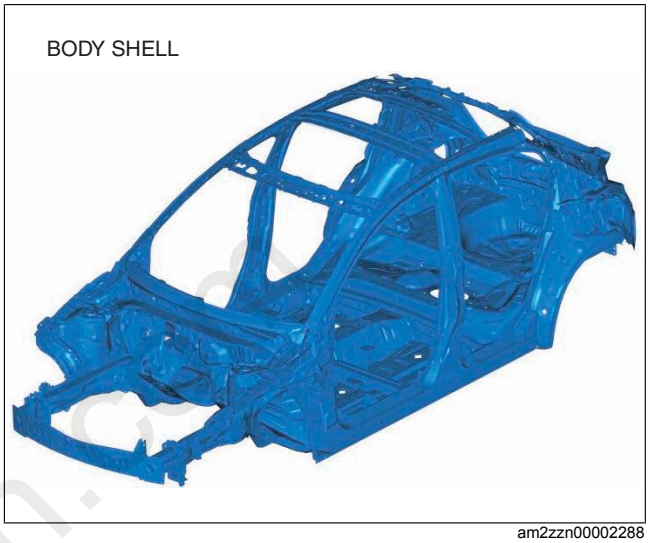
BODY SHELL

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Purpose

- The following structures have been adopted to absorb impact efficiently in the event of a collision and to suppress cabin deformation.
  - — Straight structure which achieves both high strength and weight reduction
  - — Multi-load path structure which assures a high level of collision absorption
  - — Ring structure which enhances overall rigidity
- A cross-shaped sectional front bumper reinforcement crush can and front side frames have been adopted which effectively absorb impact energy during a frontal collision and suppress cabin deformation.

Structural View

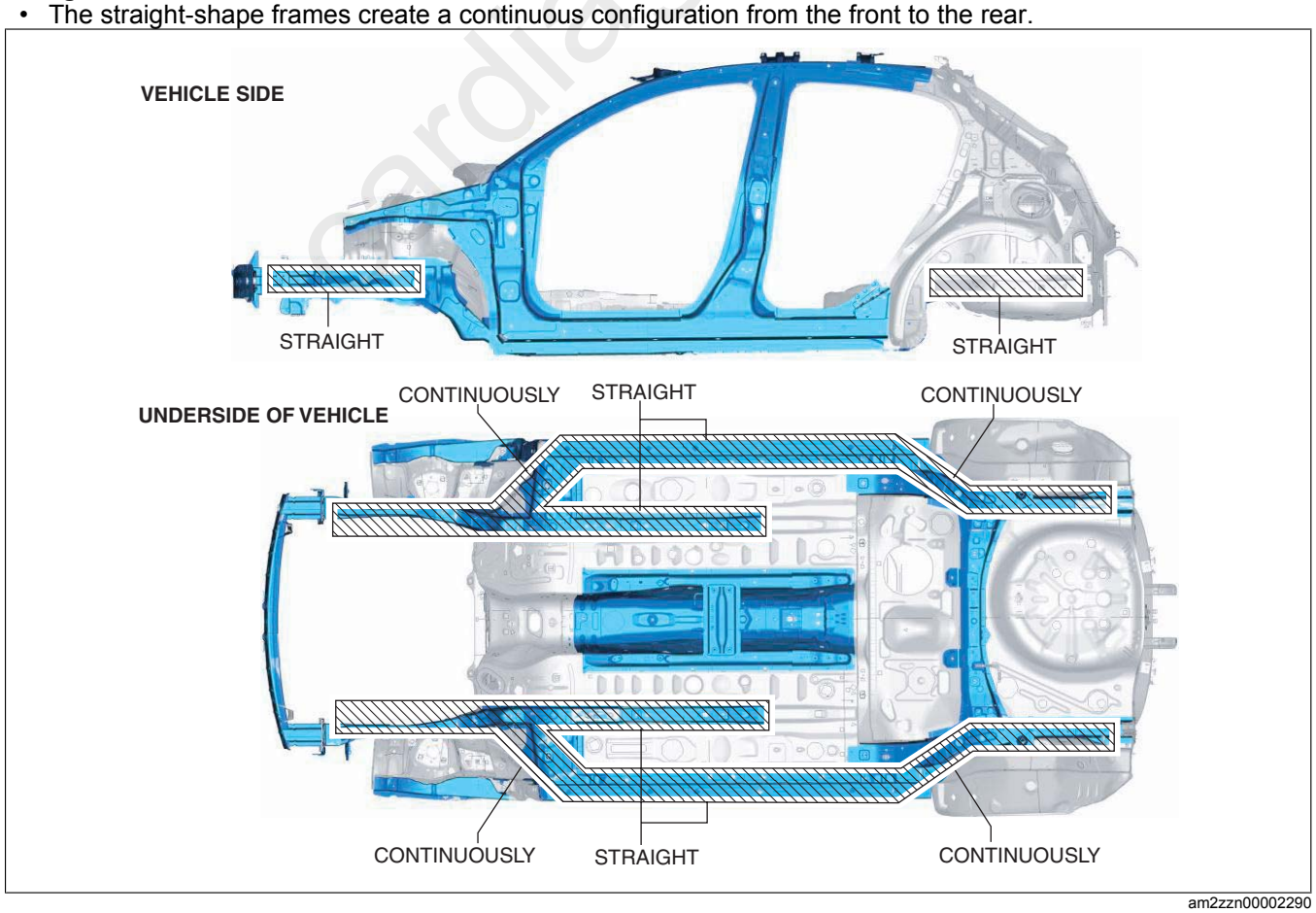


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Construction

Straight structure

- The straight-shape frames create a continuous configuration from the front to the rear.



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Multi-load path structure

- Impact energy in a collision is dispersed in multiple directions and absorbed effectively.
- To disperse and absorb frontal impact energy at the engine compartment, the following three paths have been established at the vehicle front (front frame).

  Upper path

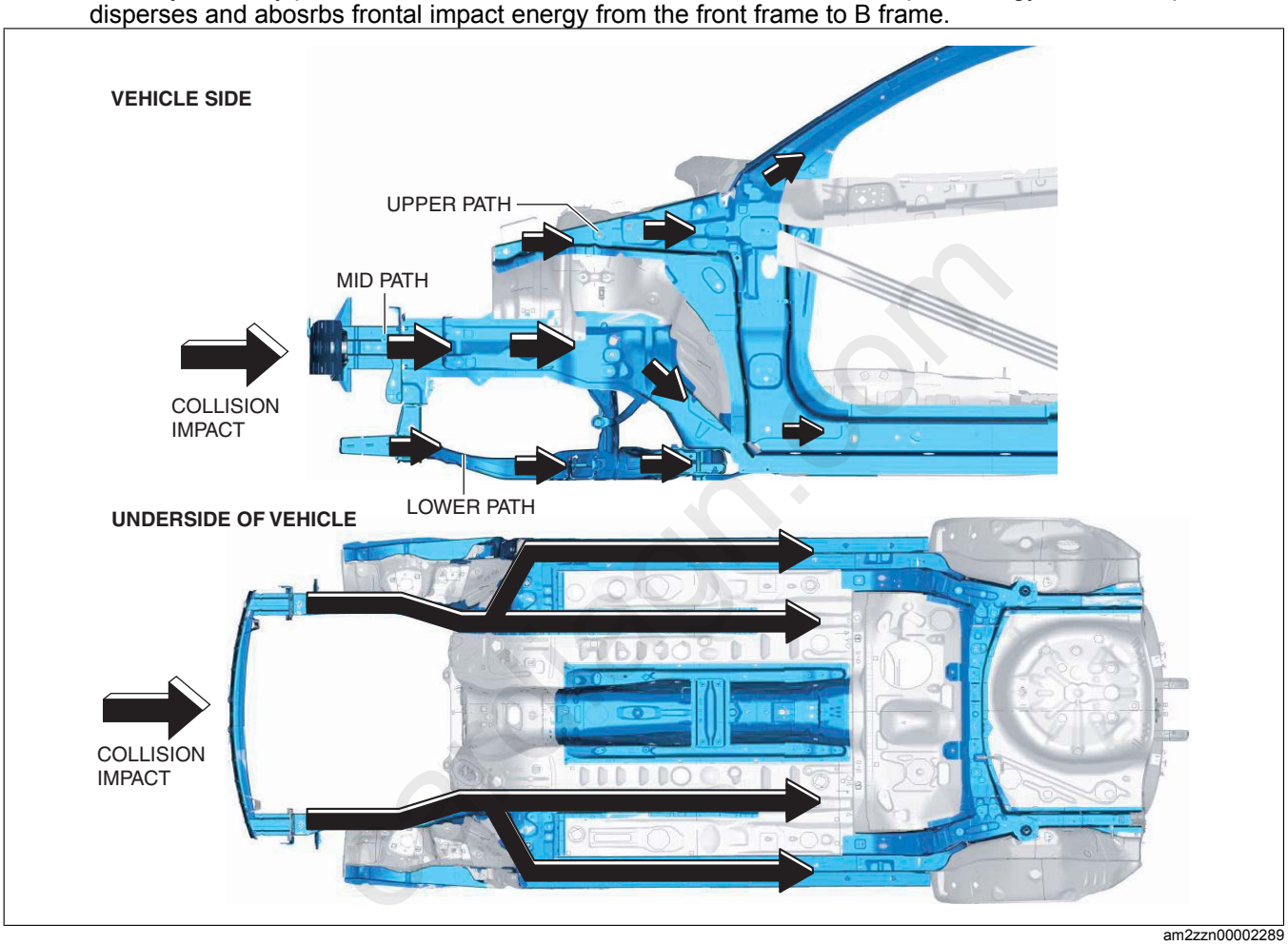
  — The upper path disperses and absorbs frontal impact energy from the front frame to A-pillar.

  Mid path

  — The mid path disperses and absorbs frontal impact energy from the front frame to the side body and floor.

  Lower path

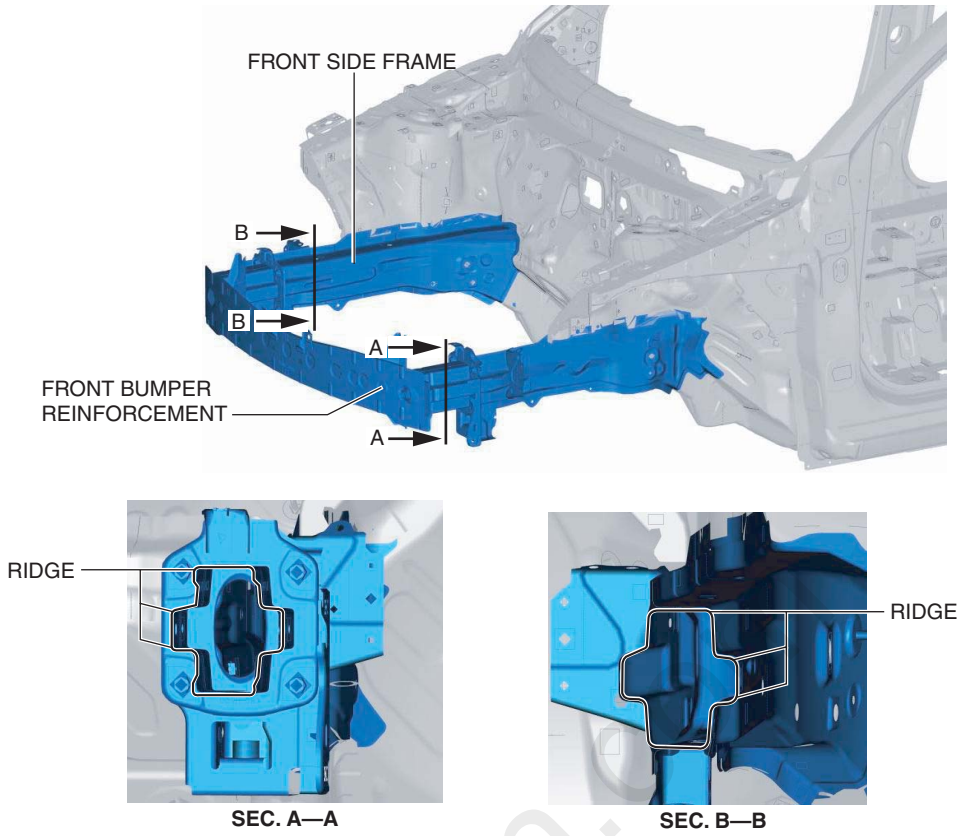
  — Not only the body parts, the front crossmember is also used to absorb impact energy. The lower path disperses and abosrbs frontal impact energy from the front frame to B frame.



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Cross sectional shape

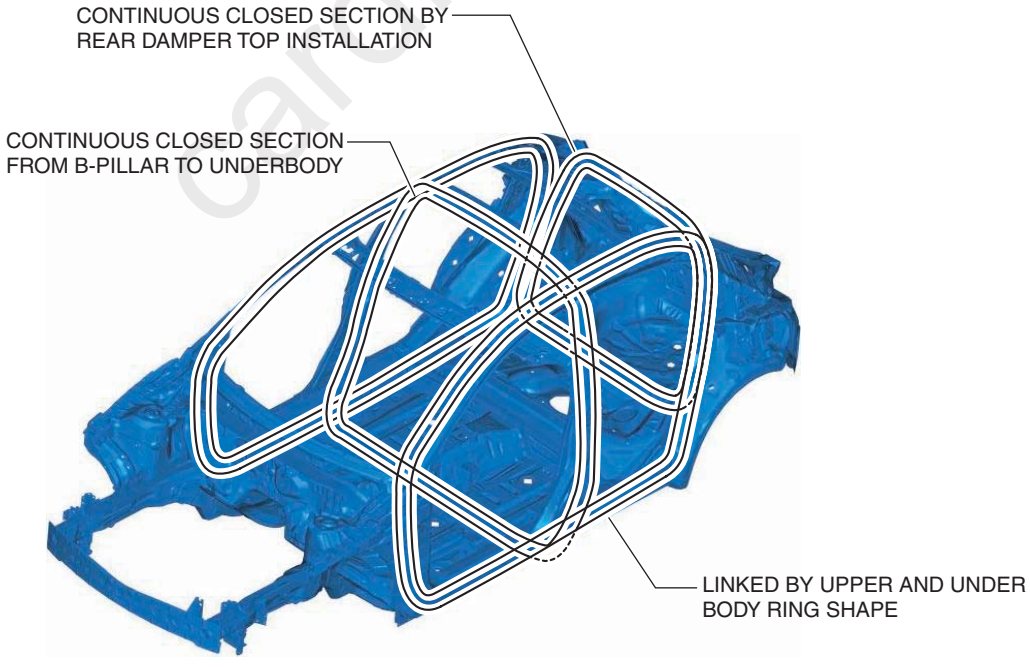
- Even using a steel plate of the same thickness, the strength can be enhanced by increasing the number of ridges. By utilizing this characteristic, the cross-section of the front bumper reinforcement crush can and front side frames is cross-shaped, achieving both high impact energy absorption and weight reduction.



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Ring structure

- Overall body rigidity has been improved using four ring structures created by the upper body including the roof rail and B-pillar combined with the underbody reinforcements.
- Cabin deformation in a collision is suppressed by creating a ring structure.



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Fail-safe

- Not applicable